

The $M \times n$ matrix $\mathcal{O}_x$ is the observability matrix with n independent columns and the $M \times M$ matrix $\mathcal{H}$ is an impulse response matrix. With these constructions of the data and definitions of matrices, the input-state-output relations can be compactly expressed as

$$\mathbf{Y} = \mathcal{O}_x \mathbf{X} + \mathcal{H} \mathbf{U}, \quad (12.167)$$

and the unknown model is contained in the matrices $\mathcal{O}_x$ and $\mathcal{H}$.

The next step is to remove the term in $\mathcal{H}$ by multiplying on the right by a matrix that is perpendicular to $\mathbf{U}$, which can be found by computing the **singular-value decomposition** (SVD) of $\mathbf{U}$ partitioned as follows

$$\mathbf{U} = [\mathbf{P}_{u1} \quad \mathbf{P}_{u2}] [\boldsymbol{\Sigma}_u \quad \mathbf{0}] \begin{bmatrix} \mathbf{Q}_{u1}^T \\ \mathbf{Q}_{u2}^T \end{bmatrix} = \mathbf{P}_u \boldsymbol{\Sigma}_u \mathbf{Q}_{u1}^T. \quad (12.168)$$

It is a property of the SVD that the matrices $\mathbf{P}_u$ and $\mathbf{Q}_u$ are orthogonal matrices and consequently that the product $\mathbf{Q}_{u1}^T \mathbf{Q}_{u2} = \mathbf{0}$. Thus if we multiply Eq. (12.167) on the right by $\mathbf{Q}_{u2}$ it is reduced to

$$\mathbf{Y} \mathbf{Q}_{u2} = \mathcal{O}_x \mathbf{X} \mathbf{Q}_{u2}. \quad (12.169)$$

The matrix on the right side of Eq. (12.169) is a combination of the columns of the observability matrix in the space of the state $\mathbf{x}$ and has n independent columns. These can be identified by again taking an SVD, this time of $\mathbf{Y} \mathbf{Q}_{u2}$ by computing matrices $\mathbf{P}$, $\mathbf{S}$, and $\mathbf{Q}$ so that

$$\mathbf{Y} \mathbf{Q}_{u2} = \mathbf{P} \mathbf{S} \mathbf{Q}^T = [\mathbf{P}_1 \quad \mathbf{P}_2] \begin{bmatrix} \boldsymbol{\Sigma} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{Q}_1^T \\ \mathbf{Q}_2^T \end{bmatrix}. \quad (12.170)$$

The number of independent columns in this product are the columns of $\mathbf{P}_1$ and equals the number of nonzero singular values in $\mathbf{S}$ which is the size of $\boldsymbol{\Sigma}$ which is therefore by definition $n \times n$. Comparing this result with Eq. (12.169) means that we can take $\mathbf{P}_1 = \mathcal{O}_x$ from which we can solve for $\mathbf{H}$ and Φ . If we wish to consider another state than $\mathbf{x}$ via a transformation $\mathbf{T}$ according to $\mathbf{x} = \mathbf{T} \mathbf{z}$ we would have $\mathcal{O}_z = \mathcal{O}_x \mathbf{T}$ and $\mathbf{P}_1 = \mathcal{O}_z \mathbf{T}^{-1}$.

With $\mathcal{O}_x$ given by Eq. (12.165), the matrix $\mathbf{H}$ is the first row.¹⁷ We next solve for Φ using the properties of $\mathcal{O}_x$ given by Eq. (12.165). If we drop the last row of $\mathcal{O}_x$ and call that reduced matrix $\overline{\mathcal{O}}_x$ and drop the first row and call that $\underline{\mathcal{O}}_x$ then it is clear that $\underline{\mathcal{O}}_x = \overline{\mathcal{O}}_x \Phi$ from which we can solve for

$$\Phi = [\overline{\mathcal{O}}_x]^\dagger \underline{\mathcal{O}}_x, \quad (12.171)$$

where $[\cdot]^\dagger$ represents the pseudo inverse and computes a least squares solution.¹⁸

¹⁷ In the multivariable case, we'd take $\mathbf{H}$ as the first n_i rows.

¹⁸ Some authors suggest using total least squares rather than ordinary least squares, as both $\underline{\mathcal{O}}_x$ and $\overline{\mathcal{O}}_x$ are subject to noise and uncertainty in real cases.

Having $\mathbf{H}$ and Φ , we now turn to computing Γ and J by operating on Eq. (12.167) to isolate $\mathcal{H}$. To do this we multiply Eq. (12.167) on the right by the pseudo inverse $\mathbf{U}^\dagger$ (which is readily found from Eq. (12.168)) and on the left by $\mathbf{P}_{u2}^T$. Since by definition $\mathbf{U}\mathbf{U}^\dagger = \mathbf{I}$ and $\mathbf{P}_{u2}^T \mathcal{O}_x = \mathbf{P}_{u2}^T \mathbf{P}_{u1} = \mathbf{0}$, these operations reduce the equation to

$$\mathbf{P}_{u2}^T \mathbf{Y} \mathbf{U}^\dagger = \mathbf{P}_{u2}^T \mathcal{H}. \quad (12.172)$$

Because Γ and J appear on the right in $\mathcal{H}$ in a linear fashion, we can solve for them by rearranging the terms in the equation. We partition the matrices in this equation consistent with $\mathcal{H}$ and write it as

$$\begin{bmatrix} K_1 & K_2 & \cdots & K_M \end{bmatrix} = \begin{bmatrix} p_1 & p_2 & \cdots & p_M \end{bmatrix} \begin{bmatrix} J & 0 & \cdots & 0 \\ \mathbf{H}\Gamma & J & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{H}\Phi^{M-2}\Gamma & \cdots & \mathbf{H}\Gamma & J \end{bmatrix}. \quad (12.173)$$

To isolate the unknowns J and Γ these equations can be reorganized and written as

$$\begin{bmatrix} K_1 \\ K_2 \\ \vdots \\ K_M \end{bmatrix} = \begin{bmatrix} p_1 & p_2 & \cdots & p_M \\ p_2 & \cdots & p_M & 0 \\ \vdots & \cdots & 0 & \vdots \\ p_M & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \mathcal{O}_x \end{bmatrix} \begin{bmatrix} J \\ \Gamma \end{bmatrix}. \quad (12.174)$$

From this equation, one can solve for J and Γ by, for example, least squares again. This completes one variation of the prediction subspace identification algorithm. Clearly there are a substantial number of choices the engineer needs to make, including the selection of M and the selection of least squares or total least squares to solve Eq. (12.171) for Φ and Eq. (12.174) for Γ and J . Perhaps more important than these decisions is the choice of n from Eq. (12.170). As given, we have ignored the possibility of noise which is always present in any practical case. Thus in practice the dimension of Σ in Eq. (12.170) is not obvious but must be selected as that value such that the *significant* singular values are kept and the negligible singular values are dropped. How clearly this choice is depends on the particular case and the other choices made in the algorithm and can be quite difficult. In important cases, models based on the prediction error method as well as one based on subspace methods are frequently computed and tested against new data and physical knowledge to guide final model selection.

12.9 Summary

In this chapter we have introduced some of the concepts of identification of dynamic systems for the purposes of control design. The main points were

- Data to be used for identification should be filtered to reduce noise and to remove known components of the transfer function.
- A good frequency response model can be computed using sinusoidal inputs, one frequency at a time.
- A chirp signal input coupled with the FFT permits one to obtain a frequency response model in one pass.
- Either batch or recursive least squares gives a good model when the equation error is an independent white noise process.
- Prediction error methods based on the maximum likelihood are among the best general methods for identification.
- State subspace-based methods estimate state realizations directly and are especially effective for multi-input-multi-output systems.

12.10 Problems

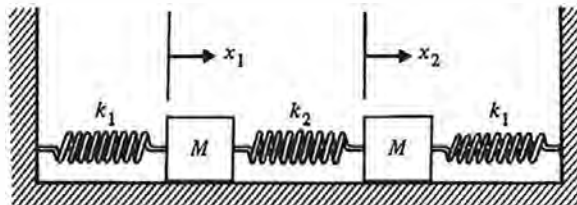
12.1 Data from a mass moving in a plane (x, y) without force are given as

0	-0.426	1.501
1	0.884	1.777
2	2.414	2.530
3	2.964	3.441
4	3.550	3.762
5	5.270	4.328
6	5.625	4.016
7	7.188	5.368
8	8.129	5.736
9	9.225	6.499
10	10.388	6.233

Plot the points and use least squares to estimate the initial value of x , the initial value of y , and the velocity of the mass in x and in y . Draw the curve (line) that corresponds to your estimates.

12.2 A simple mechanical system without friction is sketched in Fig. 12.17.

Figure 12.17
A mechanical system
used in Problem 12.2



- (a) Show that this system is described by the equations

$$\begin{aligned} M\ddot{x}_1 + k_1x_1 + k_2(x_1 - x_2) &= 0, & x_1(0) &= x_{10}, \\ M\ddot{x}_2 + k_1x_2 + k_2(x_2 - x_1) &= 0, & x_2(0) &= x_{20}. \end{aligned}$$

- (b) We wish to estimate the parameters k_1 , k_2 , and M from measurements of $x_1(t)$. However, note from (a) that if we divide by M in each equation the only identifiable parameters are $k_1/M = K_1$ and $k_2/M = K_2$. Give an intuitive argument to the effect that, if $x_{10} = x_{20}$ or if $x_{10} = -x_{20}$, it is impossible to estimate both K_1 and K_2 from $x_1(t)$. (*Hint*: Rewrite the equations using $y = x_1 - x_2$ and $z = x_1 + x_2$ as the position variables.) Show that if $x_{10} = 1$ and $x_{20} = 0$, it should be possible to estimate both K_1 and K_2 from $x_1(t)$. What do you conclude makes a "good" initial condition?
- (c) Compute a discrete equivalent system for sampling at 0.5 sec with an input that can set an initial condition on x_1 . Select $K_1 = 0.4$ and $K_2 = 0.6$. Compute the transform of X_1 if $x_{10} = 1$

$$X_1(z) = \frac{b_0z^4 + b_1z^3 + b_2z^2 + b_3z + b_4}{z^4 + a_1z^3 + a_2z^2 + a_3z + a_4},$$

where the parameters a_i are functions of K_1 , K_2 , and sampling period T .

- (d) Formulate the least-squares problem to estimate the a_i from noisy measurements of $x_1(kT)$. Set up the problem so that it is possible to ignore the b_i .
- (e) Simulate the equations of part (a) for $K_1 = 0.4$ and $K_2 = 0.6$, sample x_1 at a rate of 0.5 sec for 30 samples, and compute $\hat{\theta} = [\hat{a}_1 \ \hat{a}_2 \ \hat{a}_3 \ \hat{a}_4]^T$. Assume all the noise comes from an A/D converter operating at 10 bits including sign. If the simulation is done on a digital computer, add the appropriate noise according to the noise model of quantization discussed in Chapter 10. Compute the predicted variance in the estimate of $\hat{\theta}$.
- (f) Deliberately add additional noise of known variance to your data, recompute $\hat{\theta}$ 50 times from different runs of noisy data, and compare the sample variance of $\hat{\theta}$ to the theoretical value based on P .
- (g) Keep the number of samples fixed at 30 but vary the sample rate from 0.1 sec to 1 sec in 0.1 steps and compute

$$\frac{1}{4} \sum_{k=1}^4 \left| \frac{\hat{\theta}_k - \theta_k^0}{\theta_k^0} \right|$$

for each sample period as a measure of total estimate accuracy. What do you conclude respecting selection of sample period for identification?

- (h) Keep the sample period fixed at 0.5 sec, but compute the estimate for varying numbers of samples. Consider at least 5, 10, 30, 100, and 300 samples. Compare the same criterion used in part (g) and give an explanation of the results.

12.3 Program the recursive least-squares algorithm of Eq. (12.73) for no weighting. Use the algorithm to estimate the parameters of the system of Problem 12.2(h). Use $\hat{\theta}(5) = \mathbf{0}$ and $P(5) = \mathbf{I}$ for your initial conditions. If available, use the Identification Toolbox to repeat the exercise.

12.4 Give the transfer function $H(z)$ of the filter that corresponds to exponentially weighted least squares as a function of a and γ . Prove that $H(1) = 1$ if $a = 1 - \gamma$.

- 12.5 In Eq. (12.84) we showed that the variance in the least-squares estimate is $(A^T A)^{-1} A^T R A (A^T A)^{-1}$ and, in Eq. (12.108), we showed that the variance of the BLUE is $(A^T R^{-1} A)^{-1}$. Use the development of the proof of the BLUE following Eq. (12.103) to devise an expression for the excess variance of least squares over BLUE.
- 12.6 (Contributed by N. Gupta.) Show that the least-squares estimate $\hat{\theta} = (A^T A)^{-1} A^T Y$ results in the error squared $\mathcal{E}(Y - A\hat{\theta})^T (Y - A\hat{\theta}) = (m - n)\sigma^2$, where $\mathcal{E}e^2 = \sigma^2$, Y has m components, and $\hat{\theta}$ has n components. What estimate of σ^2 does this result suggest? *Hint:* If $M = I - A(A^T A)^{-1} A^T$, then $M^2 = M = M^T$.
- 12.7 In Eq. (12.134) we showed that the maximum likelihood estimate of σ^2 is

$$(Y - A\hat{\theta})^T (Y - A\hat{\theta}) / m.$$

Show that this estimate is biased.

- 12.8 Write a computer program to implement the search for the maximum likelihood estimate following the method of Section 12.8.
- 12.9 Simulate the system [described in Astrom and Eykhoff (1971)]
- $y_{k+1} = -a_1 y_k + b_1 u_k + v_{k+1} + c_1 v_k$, where u_k and v_k are independent sequences of unit variances and $a_1 = -0.5$, $b_1 = 1.0$, $c_1 = 0.1$.
 - Compute the least-squares estimate of a_1 and b_1 from observation of 5, 50, and 500 samples. Compare variance and bias to the theory in each case.
 - Compute the maximum-likelihood estimates of a_1 , b_1 , c_1 , and σ_v^2 from 5, 50, and 500 samples, and compare the estimates to the known true values.
- 12.10 Suppose we wish to identify a plant that is operating in a closed loop as

$$\begin{aligned} y_{k+1} &= ay_k + bu_k + e_k, \\ u_k &= -Ky_k, \end{aligned}$$

where e_k is white noise. Show that we cannot identify a and b from observation of y and u , even if K is known.

• 13 •

Nonlinear Control

Perspective on Nonlinear Control

Every physical system is nonlinear for large signals, and many systems have important nonlinearities such as friction that cannot be ignored even for small signals. Furthermore, optimal control of many systems requires a nonlinear controller. Because the variety of nonlinear systems is so vast, it is only possible here to give an introduction to the most important issues and concepts.¹ The first concern is stability, and the most important stability theory is that of Lyapunov. It is this theory that justifies the enormous attention paid to the linear case by providing the proof that, for most nonlinear systems, stability of the small signal linear approximation implies a region of stability of the nonlinear system. Beyond this result, Lyapunov theory gives us a tool that can be used to determine further regions of stability and also can guide the design of purposely nonlinear controllers in a process called Lyapunov redesign. Unfortunately, application of Lyapunov theory is often difficult and can be very frustrating. Alternative approaches, including computer simulation, heuristic methods such as the describing function, and the frequency-response-based circle criterion are also important techniques.

Design of controls for nonlinear systems may be placed in several categories. The most primitive method is to obtain a linear approximation, design a linear controller, and use computer simulation and perhaps a stability theory such as Lyapunov's to explore nonlinear behavior. In an approach known as **Lyapunov redesign** the engineer constructs a tentative Lyapunov function and deliberately designs the controller to force the system to realize the function which automatically provides a proof that the system is stable. Another widely used approach is to use feedback to reduce the equations to a linear form and continue as in

¹ In this text we assume that the model equations have unique solutions and do not exhibit the astonishing behavior of Chaos (Gleick, 1987).

the first case. This method is known in the field of robotics as the method of **computed torque**. A combination of Lyapunov redesign and computed torque that extends both has recently been developed as **backstepping**. Another approach to nonlinear design is to apply the concepts of optimal control to obtain a controller that minimizes a performance measure. A well-known case is that of minimal time control of a linear system subject to bounded controls. A practical approximation to minimal time control has been developed as **proximate time optimal systems (PTOS)**. Finally, a widely used concept is that of **adaptive control**. The fundamental idea of adaptive control is that the controller is redesigned on-line to maintain good performance in the face of a changing environment that includes major changes in the plant dynamics. Sub-categories of adaptive control are **gain scheduling**, the **model-reference adaptive control (MRAC)**, and the **self-tuning regulator (STR)**.

Chapter Overview

In this chapter, we first consider computer simulation and the heuristic analysis methods of equivalent gain and the describing function. For stability, the main analysis technique presented is the second method of Lyapunov, but also discussed are the circle criterion and the small gain theorem. An important design technique uses linear design methods based on a linear approximation to the plant model. As we will see, this approach is known as Lyapunov's first method. In Section 13.2 we consider two approaches to design by nonlinear feedback. The first is the proximate time optimal system (PTOS) design for linear systems having only saturation of the actuator as the nonlinearity, and the second is the self-tuning regulator approach to adaptive control. PTOS control is especially appropriate for motion control and adaptive control is widely used in chemical process control. In the last section of the chapter, several important approaches to computer-aided design of nonlinear systems are introduced.

**To obtain the remainder of this chapter, send a request to:
elliskagle@gmail.com**

• 14 •

Design of a Disk Drive Servo: A Case Study

Perspective on the Design Process

To illustrate many of the concepts developed in this book, the design of a disk drive head positioning servomechanism provides an excellent case study. In most textbook examples, the problems are *highly* simplified for pedagogy: Too much detail often obscures the underlying point. Indeed, most of this text has followed this principle and many of the individual problems of this case study are used to illustrate various parts of the design process in earlier chapters. However, for a final design example we will not spare all of the detail but will present some of the difficult realities of a real problem. The problem requires constructing models of the plant, the actuator, and the sensor and verifying these models with simulations. Next comes determining the specifications both for the components and for the entire system. Following the problem formulation comes the design of a small-signal linear controller and finally the design is completed with a nonlinear controller for handling large signal behavior. The last step is testing the design in simulation for its performance in the presence of varying parameters before constructing a prototype and testing performance by experiment. We begin the case study by giving a description of the disk drive which will serve to set the stage for the specification development in our design, and this too will be in more depth than in our typical examples.

Chapter Overview

A general discussion of disk drives and current performance specifications are given in Section 14.1. In Section 14.2 the several components that comprise the disk drive servo are described and models given. The specific performance

specifications for the servo case study are given in Section 14.3. Finally, the details of the design are presented in Section 14.4. As with earlier chapters, this ends with a set of problems designed to enhance the learning process.

**To obtain the remainder of this chapter, send a request to:
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♦ A ♦

Examples

In this appendix we describe five control problems which are used to illustrate the analysis and design techniques developed in the text. These are collected here in order to permit references to them at a number of places in the text without the need to repeat the derivation of the model or discuss the specifications each time the example is used.

A.1 Single-Axis Satellite Attitude Control

Satellites often require attitude control for proper orientation of antennas and sensors with respect to the earth. Figure A.1 shows a communications satellite with a three-axis attitude-control system. To gain insight into the three-axis problem we often consider one axis at a time. Figure A.2 depicts this case where motion is allowed only about an axis perpendicular to the page. The equations of motion of the system are given by

$$I\ddot{\theta} = M_c + M_d, \quad (\text{A.1})$$

where I is the moment of inertia of the satellite about its mass center, M_c is the control torque applied by the thrusters, M_d are the disturbance torques, and θ is the angle of the satellite axis with respect to an “inertial” reference. The inertial reference must have no angular acceleration. Normalizing, we define

$$u = M_c/I, \quad w_d = M_d/I \quad (\text{A.2})$$

and obtain

$$\ddot{\theta} = u + w_d. \quad (\text{A.3})$$

Figure A.1
Communications
satellite (courtesy Space
Systems-Loral)

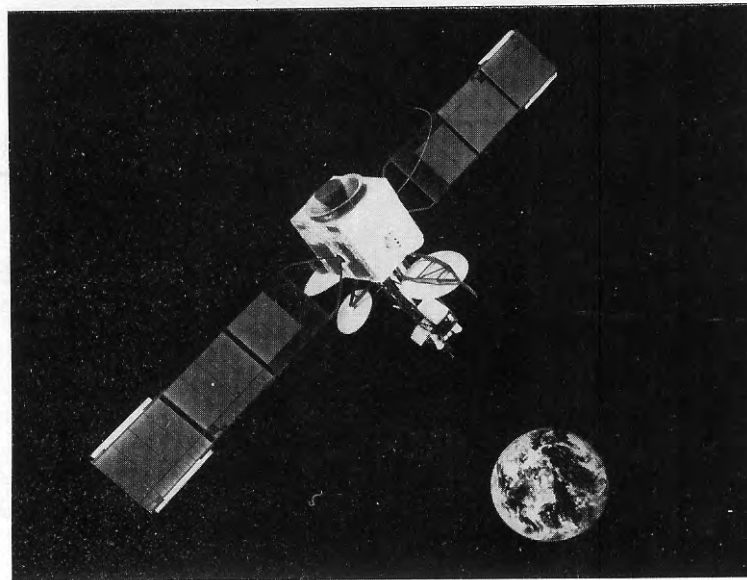
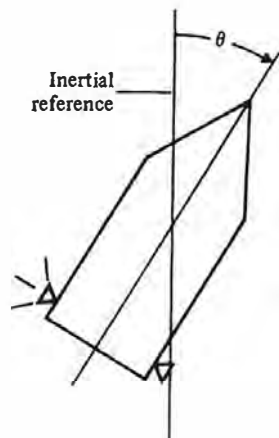


Figure A.2
Satellite-control
schematic



Taking the Laplace transform

$$\theta(s) = \frac{1}{s^2} [u(s) + w_d(s)], \quad (\text{A.4})$$

which becomes, with no disturbance

$$\frac{\theta(s)}{u(s)} = \frac{1}{s^2} = G_1(s). \quad (\text{A.5})$$

In the discrete case with u being applied through a zero-order hold, we can use the methods of Chapter 4 to obtain the discrete transfer function

$$\begin{aligned} G_1(z) &= \frac{\theta(z)}{u(z)} \\ &= \frac{T^2}{2} \frac{(z+1)}{(z-1)^2}. \end{aligned} \quad (\text{A.6})$$

A.2 A Servomechanism for Antenna Azimuth Control

It is desired to control the elevation of an antenna designed to track a satellite as shown in Fig. A.3. The antenna and drive parts have a moment of inertia J and damping B arising to some extent from bearing and aerodynamic friction, but mostly from the back emf of the DC-drive motor. A schematic is shown in Fig. A.4.

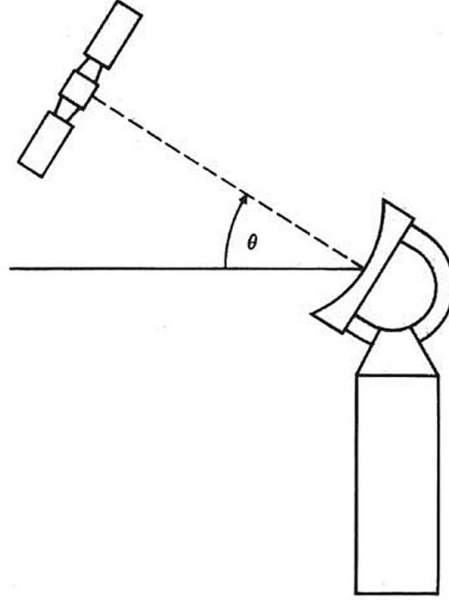
The equation of motion is

$$J\ddot{\theta} + B\dot{\theta} = T_c + T_d, \quad (\text{A.7})$$

Figure A.3
Satellite tracking
antenna (courtesy Space
Systems-Loral)



Figure A.4
Schematic diagram of
antenna



where T_c is the net torque from the drive motor and T_d is the disturbance torque due to wind. If we define

$$B/J = a, \quad u = T_c/B, \quad w_d = T_d/B,$$

the equation reduces to

$$\frac{1}{a} \ddot{\theta} + \dot{\theta} = u + w_d. \quad (\text{A.8})$$

Transformed, it is

$$\theta(s) = \frac{1}{s(s/a + 1)} [u(s) + w_d(s)] \quad (\text{A.9})$$

or, with no disturbances,

$$\frac{\theta(s)}{u(s)} = \frac{1}{s(s/a + 1)} = G_2(s). \quad (\text{A.10})$$

The discrete case with $u(k)$ applied through a zero-order hold yields

$$G_2(z) = \frac{\theta(z)}{u(z)} = K \frac{(z + b)}{(z - 1)(z - e^{-aT})}, \quad (\text{A.11})$$

where

$$K = \frac{aT - 1 + e^{-aT}}{a}, \quad b = \frac{1 - e^{-aT} - aTe^{-aT}}{aT - 1 + e^{-aT}}.$$

The model given by Eq. (A.7) is only a linear approximation to the true motion, of course. In reality there are many nonlinear effects that must be considered in the final design. To give some ideas of these considerations, we present the major features of the angle, θ_s , of the satellite to be followed and of the servomechanism that must be designed to achieve the tracking.

The general shape of $\theta_s(t)$ and its velocity and acceleration are sketched in Fig. A.5. The peak values of velocity and acceleration depend critically on the altitude of the satellite above the earth and on the elevation above the horizon of the orbit seen from the antenna. An orbit of 200 miles above the earth that passes at an elevation above the antenna of 86° requires a maximum azimuth rate of 0.34 rad/sec (about $20^\circ/\text{sec}$). For the purpose of setting the tracking accuracy and gains, a reasonable assumption is that the antenna should be capable of following this peak velocity in the steady state with acceptable error.

The size of the allowable tracking error is determined by the antenna properties. The purpose, of course, is to permit acceptable communication signals to be received by the antenna electronics. For this, we must consider the dependence of signal amplitude on pointing error. A sketch of a typical pattern is shown in Fig. A.6. The beam width $\Delta\theta$ is the range of tracking error permissible if

Figure A.5
General shape of the
command angle and its
first two derivatives,
which the tracking
antenna must follow

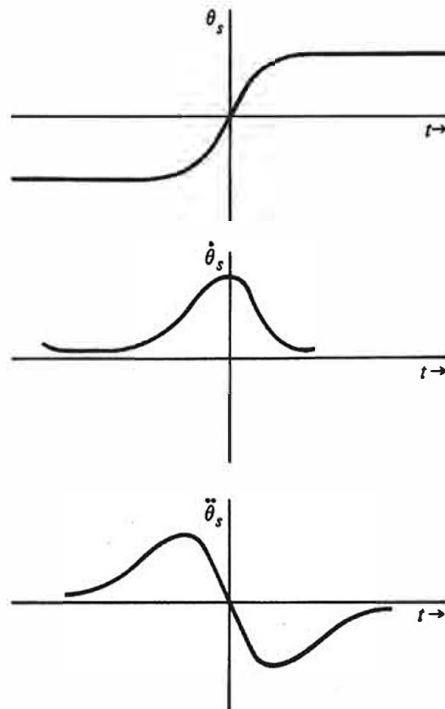
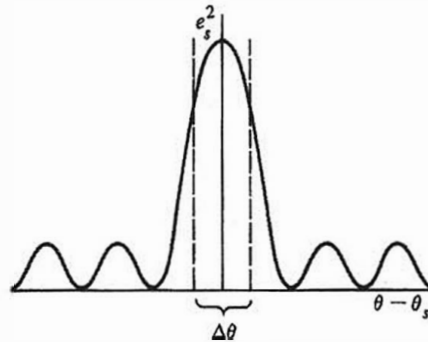


Figure A.6
Plot of signal power
received as a function of
tracking error



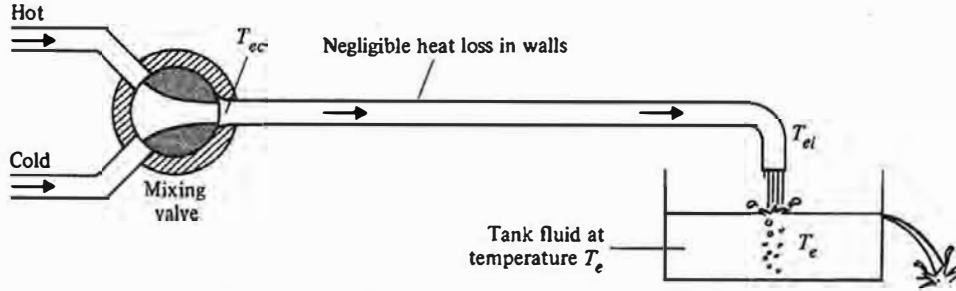
acceptable communications are to be achieved. Typical values of the beam width can vary from a few degrees (0.1 rad) to less than one degree (0.01 rad). The total error will be composed mainly of tracking errors due to θ_s motion, wind-gust errors, and random errors caused by noise in the measurement of satellite position. A reasonable allowance is to permit tracking errors of 10% of the beam width and an equal contribution from wind gusts. For a beam width of 0.1 rad, then, we allow a tracking error of 0.01 rad, from which it follows that $K_v = 1$ (if $\dot{\theta}_{\max} = 0.01$ rad/sec). For a 60-ft. tracking antenna, a typical beam width is 0.01 rad, and if 0.34 rad/sec must be followed, the velocity constant for a 10% error (0.001 rad) is $K_v = 0.34/0.001 = 340$. For a velocity constant of this magnitude, a conversion to a Type 2 system with two poles at $z = 1$ is probably required.

In addition to considerations of tracking error, the designer must take into account the fact that the drive motor has finite torque and power capability. In selecting the drive motor, one must keep in mind that the total torque must be capable of meeting the acceleration demands of tracking, overcoming the wind torque, and overcoming the static friction of the drive-train gears and antenna mount. The power of the motor must be able to supply this torque at the maximum tracking velocity.

A.3 Temperature Control of Fluid in a Tank

The temperature of a tank of fluid with a constant flow rate in and out is to be controlled by adjusting the temperature of the incoming fluid. The temperature of the incoming fluid is controlled by a mixing valve that adjusts the relative amounts of hot and cold supplies of the fluid (Fig. A.7). Because of the distance between the valve and the point of discharge into the tank, there is a time delay between the application of a change in the mixing valve and the discharge of the flow with the changed temperature into the tank.

Figure A.7
Tank temperature control



The differential equation governing the tank temperature is

$$\dot{T}_e = \frac{1}{cM} (q_{in} - q_{out}), \quad (\text{A.12})$$

where

$$\begin{aligned} T_e &= \text{tank temperature,} \\ c &= \text{specific heat of the fluid,} \\ M &= \text{fluid mass contained in the tank,} \\ q_{in} &= c\dot{m}_{in}T_{ei}, \\ q_{out} &= c\dot{m}_{out}T_e, \\ \dot{m} &= \text{mass flow rate } (= \dot{m}_{in} = \dot{m}_{out}), \text{ and} \\ T_{ei} &= \text{temperature of fluid entering tank;} \end{aligned}$$

but the temperature at the input to the tank at time t is the control temperature τ_d sec in the past, which can be expressed as

$$T_{ei}(t) = T_{ec}(t - \tau_d), \quad (\text{A.13})$$

where

$$\begin{aligned} \tau_d &= \text{delay time, and} \\ T_{ec} &= \text{temperature of fluid immediately after the control valve and directly} \\ &\quad \text{controllable by the valve.} \end{aligned}$$

Combining constants, we obtain

$$\dot{T}_e(t) + aT_e(t) = aT_{ec}(t - \tau_d), \quad (\text{A.14})$$

where

$$a = \dot{m}/M,$$

which, transformed, becomes

$$\frac{T_e(s)}{T_{ec}(s)} = \frac{e^{-\tau_d s}}{s/a + 1} = G_3(s). \quad (\text{A.15})$$

To form the discrete transfer function of G_3 preceded by a zero-order hold, we must compute

$$G_3(z) = \mathcal{Z} \left\{ \frac{1 - e^{-Ts}}{s} \frac{e^{-\tau_d s}}{s/a + 1} \right\}.$$

We assume that $\tau_d = lT - mT$, $0 \leq m < 1$. Then

$$\begin{aligned} G_3(z) &= \mathcal{Z} \left\{ \frac{1 - e^{-Ts}}{s} \frac{e^{-lTs} e^{mTs}}{s/a + 1} \right\} \\ &= (1 - z^{-1}) z^{-l} \mathcal{Z} \left\{ \frac{e^{mTs}}{s(s/a + 1)} \right\} \\ &= (1 - z^{-1}) z^{-l} \mathcal{Z} \left\{ \frac{e^{mTs}}{s} - \frac{e^{mTs}}{s + a} \right\} \\ &= \frac{z - 1}{z} \frac{1}{z^l} \mathcal{Z} \{ 1(t + mT) - e^{-a(t+mT)} 1(t + mT) \} \\ &= \frac{z - 1}{z} \frac{1}{z^l} \left\{ \frac{z}{z - 1} - \frac{e^{-amT} z}{z - e^{-aT}} \right\} \\ &= \frac{1}{z^l} \frac{(1 - e^{-amT})z + e^{-amT} - e^{-aT}}{z - e^{-aT}} \\ &= \frac{1 - e^{-amT}}{z^l} \frac{z + \alpha}{z - e^{-aT}}; \\ \alpha &= \frac{e^{-amT} - e^{-aT}}{1 - e^{-amT}}. \end{aligned} \quad (\text{A.16})$$

From Eq. (A.16) it is easy to see that the zero location, $-\alpha$, varies from $\alpha = \infty$ at $m = 0$ to $\alpha = 0$ as $m \rightarrow 1$ and that $G_3(1) = 1.0$ for all a , m , and l .

For the specific values of $\tau_d = 1.5$, $T = 1$, $a = 1$, Eq. (A.16) reduces to ($l = 2$, $m = \frac{1}{2}$)

$$G_3(z) = \frac{z + 0.6065}{z^2(z - 0.3679)}.$$

A.4 Control Through a Flexible Structure

Many controlled systems have some structural flexibility in some portion of the system. The spacecraft of Fig. A.1 may not be perfectly rigid, the angle tracker of Fig. A.3 may have some flexibility between the angle observed by the antenna and the angle of the base, and almost any mechanical system such as a robot arm or a disk drive read/write head assembly would exhibit some degree of structural flexibility.

Conceptually, these systems are equivalent to the double mass–spring device shown in Fig. A.8. The equations of motion are

$$\begin{aligned} M\ddot{y} + (\dot{y} - \dot{d})b + (y - d)k &= u, \\ m\ddot{d} + (\dot{d} - \dot{y})b + (d - y)k &= 0, \end{aligned} \quad (\text{A.17})$$

which, when transformed, become,

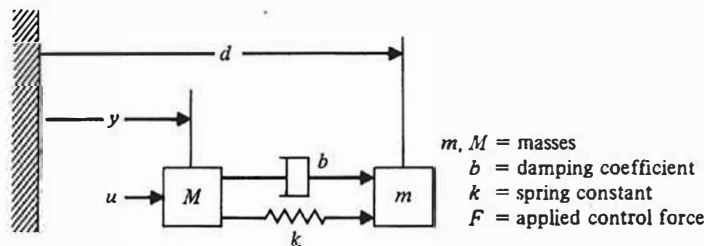
$$\begin{bmatrix} Ms^2 + bs + k & -(bs + k) \\ -(bs + k) & ms^2 + bs + k \end{bmatrix} \begin{bmatrix} y(s) \\ d(s) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(s). \quad (\text{A.18})$$

Two transfer functions are of interest: one between u and d , the other between u and y . The first represents the case where there is structural flexibility between the sensor and the actuator, a situation called the noncollocated case. The second represents the collocated case where the sensor is placed on the same rigid body as the actuator, but there exists a mechanical oscillation elsewhere in the system that is coupled to the mass to which the actuator and sensor are attached.

Using Eq. (A.18) we can obtain both transfer functions

$$\frac{d(s)}{u(s)} = \frac{1}{M} \frac{\left(\frac{b}{m}s + \frac{k}{m}\right)}{s^2 \left[s^2 + \left(1 + \frac{m}{M}\right) \left(\frac{b}{m}s + \frac{k}{m}\right) \right]}, \quad (\text{A.19})$$

Figure A.8
Double-mass spring
system



which becomes for the typical case with very low damping

$$\frac{d(s)}{u(s)} \approx \frac{1}{M} \frac{k/m}{s^2 \left[s^2 + \left(1 + \frac{m}{M} \right) \left(\frac{b}{m}s + \frac{k}{m} \right) \right]} = G_4(s) \quad (\text{A.20})$$

and

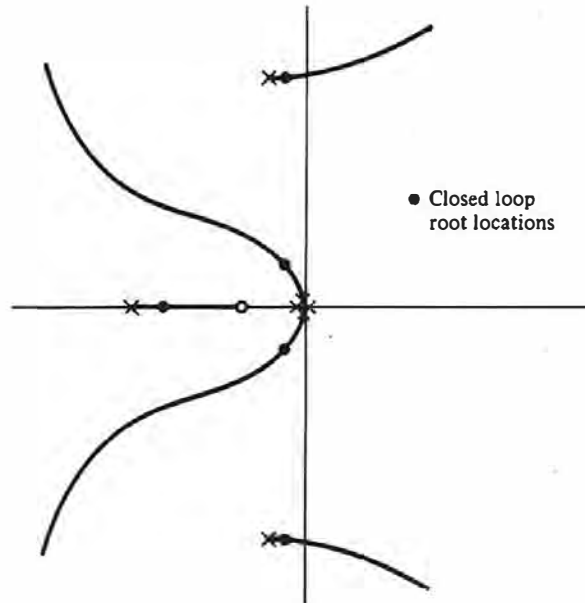
$$\frac{y(s)}{u(s)} = \frac{1}{M} \frac{\left(s^2 + \frac{b}{m}s + \frac{k}{m} \right)}{s^2 \left[s^2 + \left(1 + \frac{m}{M} \right) \left(\frac{b}{m}s + \frac{k}{m} \right) \right]} = G_5(s). \quad (\text{A.21})$$

For $m/M \ll 1$, Eq. (A.21) indicates that the system dynamics are essentially $1/s^2$, but there are poles and zeros at the structural bending-mode frequency that almost cancel each other. A small value of m/M in this simplified example represents the fact that the flexibility is not a dominant response to the control input u and $1/s^2$ is a good model for many such problems.

Equation (A.20) always has the $1/s^2$ ("rigid-body") poles plus the resonance poles without the neighboring zeros of Eq. (A.21). The noncollocated case is the harder one to control for if a lead compensation is used to stabilize Eq. (A.20), the structural mode is destabilized, as shown in the root locus of Fig. A.9, and depending on the feedback gain and original structural damping, the structural mode could well be driven unstable.

Figure A.9

Root locus for double mass-spring system with flexibility between sensor and actuator. The noncollocated case



A.5 Control of a Pressurized Flow Box

This example is multivariable because it has two control variables and two measurable states. The basic function of the system is to keep the paper-stock flow out the bottom opening (slice) (Fig. A.10) at a consistent rate. The differential equations of motion are¹

$$\begin{bmatrix} \dot{H} \\ \dot{h} \\ \dot{u}_a \end{bmatrix} = \begin{bmatrix} -0.2 & +0.1 & +1 \\ -0.05 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} H \\ h \\ u_a \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0.7 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} u_c \\ u_s \end{bmatrix}, \quad (\text{A.22})$$

where

$H = h + p/\rho_s g$ = total head perturbation,

p = air pressure,

$\bar{p}$ = mean air pressure,

h = stock-level perturbation,

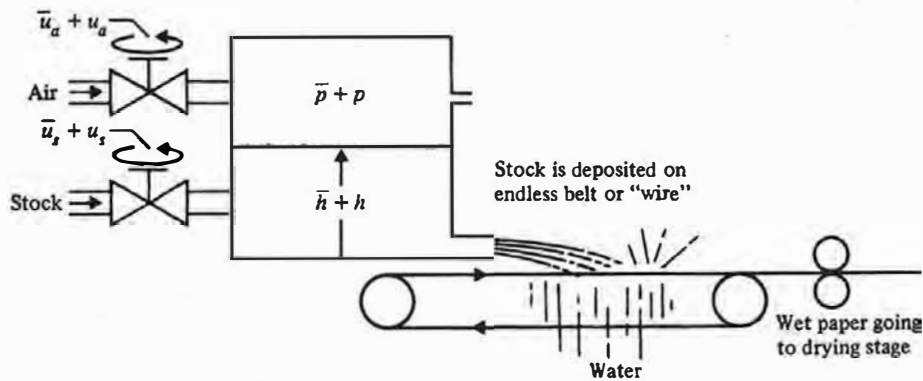
u_a = perturbation in air-valve opening,

u_c = command value to air valve, and

u_s = perturbation in stock-valve opening.

Note from Eq. (A.22) that there is two-way coupling between the two outputs, H and h , and that u_s affects both states; therefore, it is difficult to decouple the system so as to influence either H or h alone with a single control. The third

Figure A.10
Pressurized flow box



¹ These numbers are mainly contrived and might not correspond to any real head box, alive or dead.

equation in Eq. (A.22) represents dynamics of the air-valve actuator, which is such that u_a does not respond instantaneously to an air command u_c . The transfer functions between u_c and u_s and H and h are

$$G_6(s) = \frac{1}{\Delta(s)} \begin{bmatrix} s & (s+1)(s+0.07) \\ -0.05 & 0.7(s+1)(s+0.13) \end{bmatrix}$$

where $\Delta(s) = (s+1)(s+0.1707)(s+0.02929)$.

♦ B ♦

Tables

B.1 Properties of z-Transforms

Let $\mathcal{F}_i(s)$ be the Laplace transform of $f_i(t)$ and $F_i(z)$ be the z-transform of $f_i(kT)$

Table B.1

Number	Laplace Transform	Samples	z-Transform	Comment
—	$\mathcal{F}_i(s)$	$f_i(kT)$	$F_i(z)$	
1	$\alpha \mathcal{F}_1(s) + \beta \mathcal{F}_2(s)$	$\alpha f_1(kT) + \beta f_2(kT)$	$\alpha F_1(z) + \beta F_2(z)$	The z-transform is linear
2	$\mathcal{F}_1(e^{Ts})\mathcal{F}_2(s)$	$\sum_{\ell=-\infty}^{\infty} f_1(\ell T)f_2(kT - \ell T)$	$F_1(z)F_2(z)$	Discrete convolution corresponds to product of z-transforms
3	$e^{+nTs}\mathcal{F}(s)$	$f(kT + nT)$	$z^n F(z)$	Shift in time
4	$\mathcal{F}(s + a)$	$e^{-akT}f(kT)$	$F(e^{aT}z)$	Shift in frequency
5	—	$\lim_{k \rightarrow \infty} f(kT)$	$\lim_{z \rightarrow 1} (z - 1)F(z)$	If all poles of $(z - 1)F(z)$ are inside the unit circle and $F(z)$ converges for $1 \leq z $
6	$\mathcal{F}(s/\omega_n)$	$f(\omega_n kT)$	$F(z; \omega_n T)$	Time and frequency scaling
7	—	$f_1(kT)f_2(kT)$	$\frac{1}{2\pi j} \oint_{c_3} F_1(\xi)F_2(z/\xi) \frac{d\xi}{\xi}$	Time product
8	$\mathcal{F}_3(s) = \mathcal{F}_1(s)\mathcal{F}_2(s)$	$\int_{-\infty}^{\infty} f_1(\tau)f_2(kT - \tau)d\tau$	$F_3(z)$	Continuous convolution does <i>not</i> correspond to product of z-transforms

B.2 Table of z-Transforms

$\mathcal{F}(s)$ is the Laplace transform of $f(t)$ and $F(z)$ is the z-transform of $f(kT)$. Unless otherwise noted, $f(t) = 0$, $t < 0$ and the region of convergence of $F(z)$ is outside a circle $r < |z|$ such that all poles of $F(z)$ are inside r .

Table B.2

Number	$\mathcal{F}(s)$	$f(kT)$	$F(z)$
1	—	1, $k = 0$; 0, $k \neq 0$	1
2	—	1, $k = m$; 0, $k \neq m$	z^{-m}
3	$\frac{1}{s}$	1(kT)	$\frac{z}{z-1}$
4	$\frac{1}{s^2}$	kT	$\frac{Tz}{(z-1)^2}$
5	$\frac{1}{s^3}$	$\frac{1}{2!}(kT)^2$	$\frac{T^2 z(z+1)}{2(z-1)^3}$
6	$\frac{1}{s^4}$	$\frac{1}{3!}(kT)^3$	$\frac{T^3 z(z^2+4z+1)}{6(z-1)^4}$
7	$\frac{1}{s^m}$	$\lim_{a \rightarrow 0} \frac{(-1)^{m-1}}{(m-1)!} \frac{\partial^{m-1}}{\partial a^{m-1}} e^{-akT}$	$\lim_{a \rightarrow 0} \frac{(-1)^{m-1}}{(m-1)!} \frac{\partial^{m-1}}{\partial a^{m-1}} \frac{z}{z - e^{-aT}}$
8	$\frac{1}{s+a}$	e^{-akT}	$\frac{z}{z - e^{-aT}}$
9	$\frac{1}{(s+a)^2}$	$kT e^{-akT}$	$\frac{Tze^{-aT}}{(z - e^{-aT})^2}$
10	$\frac{1}{(s+a)^3}$	$\frac{1}{2}(kT)^2 e^{-akT}$	$\frac{T^2 e^{-aT} z(z + e^{-aT})}{2(z - e^{-aT})^3}$
11	$\frac{1}{(s+a)^m}$	$\frac{(-1)^{m-1}}{(m-1)!} \frac{\partial^{m-1}}{\partial a^{m-1}} (e^{-akT})$	$\frac{(-1)^{m-1}}{(m-1)!} \frac{\partial^{m-1}}{\partial a^{m-1}} \frac{z}{z - e^{-aT}}$
12	$\frac{a}{s(s+a)}$	$1 - e^{-akT}$	$\frac{z(1 - e^{-aT})}{(z-1)(z - e^{-aT})}$

Table B.2

Number	$\mathcal{F}(s)$	$f(kT)$	$F(z)$
13	$\frac{a}{s^2(s+a)}$	$\frac{1}{a}(akT - 1 + e^{-akT})$	$\frac{z[(aT - 1 + e^{-aT})z + (1 - e^{-aT} - aTe^{-aT})]}{a(z-1)^2(z - e^{-aT})}$
14	$\frac{b-a}{(s+a)(s+b)}$	$(e^{-akT} - e^{-bkT})$	$\frac{(e^{-aT} - e^{-bT})z}{(z - e^{-aT})(z - e^{-bT})}$
15	$\frac{s}{(s+a)^2}$	$(1 - akT)e^{-akT}$	$\frac{z[z - e^{-aT}(1 + aT)]}{(z - e^{-aT})^2}$
16	$\frac{a^2}{s(s+a)^2}$	$1 - e^{-akT}(1 + akT)$	$\frac{z[z(1 - e^{-aT} - aTe^{-aT}) + e^{-2aT} - e^{-aT} + aTe^{-aT}]}{(z-1)(z - e^{-aT})^2}$
17	$\frac{(b-a)s}{(s+a)(s+b)}$	$be^{-bkT} - ae^{-akT}$	$\frac{z[z(b-a) - (be^{-aT} - ae^{-bT})]}{(z - e^{-aT})(z - e^{-bT})}$
18	$\frac{a}{s^2 + a^2}$	$\sin akT$	$\frac{z \sin aT}{z^2 - (2 \cos aT)z + 1}$
19	$\frac{s}{s^2 + a^2}$	$\cos akT$	$\frac{z(z - \cos aT)}{z^2 - (2 \cos aT)z + 1}$
20	$\frac{s+a}{(s+a)^2 + b^2}$	$e^{-akT} \cos bkT$	$\frac{z(z - e^{-aT} \cos bT)}{z^2 - 2e^{-aT}(\cos bT)z + e^{-2aT}}$
21	$\frac{b}{(s+a)^2 + b^2}$	$e^{-akT} \sin bkT$	$\frac{ze^{-aT} \sin bT}{z^2 - 2e^{-aT}(\cos bT)z + e^{-2aT}}$
22	$\frac{a^2 + b^2}{s((s+a)^2 + b^2)}$	$1 - e^{-akT} \left(\cos bkT + \frac{a}{b} \sin bkT \right)$	$\frac{z(Az + B)}{(z-1)(z^2 - 2e^{-aT}(\cos bT)z + e^{-2aT})}$ $A = 1 - e^{-aT} \cos bT - \frac{a}{b} e^{-aT} \sin bT$ $B = e^{-2aT} + \frac{a}{b} e^{-aT} \sin bT - e^{-aT} \cos bT$



A Few Results from Matrix Analysis

Although we assume the reader has some acquaintance with linear equations and determinants, there are a few results of a more advanced character that even elementary control-system theory requires, and these are collected here for reference in the text. For further study, a good choice is Strang (1976).

C.1 Determinants and the Matrix Inverse

The determinant of a product of two square matrices is the product of their determinants

$$\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}. \quad (\text{C.1})$$

If a matrix is diagonal, then the determinant is the product of the elements on the diagonal.

If the matrix is partitioned with square elements on the main diagonal, then an extension of this result applies, namely,

$$\det \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{B} & \mathbf{C} \end{bmatrix} = \det \mathbf{A} \det \mathbf{C}, \quad \text{if } \mathbf{A} \text{ and } \mathbf{C} \text{ are square submatrices.} \quad (\text{C.2})$$

Suppose $\mathbf{A}$ is a matrix of dimensions $m \times n$ and $\mathbf{B}$ is of dimension $n \times m$. Let $\mathbf{I}_m$ and $\mathbf{I}_n$ be the identity matrices of size $m \times m$ and $n \times n$, respectively. Then

$$\det [\mathbf{I}_n + \mathbf{BA}] = \det [\mathbf{I}_m + \mathbf{AB}]. \quad (\text{C.3})$$

To show this result, we consider the determinant of the matrix product

$$\det \begin{bmatrix} \mathbf{I}_m & \mathbf{0} \\ \mathbf{B} & \mathbf{I}_n \end{bmatrix} \begin{bmatrix} \mathbf{I}_m & \mathbf{A} \\ -\mathbf{B} & \mathbf{I}_n \end{bmatrix} = \det \begin{bmatrix} \mathbf{I}_m & \mathbf{A} \\ \mathbf{0} & \mathbf{I}_n + \mathbf{BA} \end{bmatrix} = \det [\mathbf{I}_n + \mathbf{BA}].$$

But this is also equal to

$$\det \begin{bmatrix} \mathbf{I}_m & -\mathbf{A} \\ \mathbf{0} & \mathbf{I}_n \end{bmatrix} \det \begin{bmatrix} \mathbf{I}_m & \mathbf{A} \\ -\mathbf{B} & \mathbf{I}_n \end{bmatrix} = \det \begin{bmatrix} \mathbf{I}_m + \mathbf{AB} & \mathbf{0} \\ -\mathbf{B} & \mathbf{I}_n \end{bmatrix} = \det [\mathbf{I}_m + \mathbf{AB}],$$

and therefore these two determinants are equal to each other, which is Eq. (C.3).

If the determinant of a matrix $\mathbf{A}$ is not zero, then we can define a related matrix $\mathbf{A}^{-1}$, called " $\mathbf{A}$ inverse," which has the property that

$$\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}. \quad (\text{C.4})$$

According to property Eq. (C.1) we have

$$\det \mathbf{AA}^{-1} = \det \mathbf{A} \cdot \det \mathbf{A}^{-1} = 1,$$

or

$$\det \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}.$$

It can be shown that there is an $n \times n$ matrix called the **adjugate** of $\mathbf{A}$ with elements composed of sums of products of the elements of $\mathbf{A}^1$ and having the property that

$$\mathbf{A} \cdot \text{adj } \mathbf{A} = \det \mathbf{A} \cdot \mathbf{I}. \quad (\text{C.5})$$

Thus, if the determinant of $\mathbf{A}$ is not zero, the inverse of $\mathbf{A}$ is given by

$$\mathbf{A}^{-1} = \frac{\text{adj } \mathbf{A}}{\det \mathbf{A}}.$$

A famous and useful formula for the inverse of a combination of matrices has come to be called the **matrix inversion lemma** in the control literature. It arises in the development of recursive algorithms for estimation, as found in Chapter 9. The formula is as follows: If $\det \mathbf{A}$, $\det \mathbf{C}$, and $\det (\mathbf{A} + \mathbf{BCD})$ are different from zero, then we have the matrix inversion lemma:

$$(\mathbf{A} + \mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1}. \quad (\text{C.6})$$

The truth of Eq. (C.6) is readily confirmed if we multiply both sides by $\mathbf{A} + \mathbf{BCD}$ to obtain

$$\begin{aligned} \mathbf{I} &= \mathbf{I} + \mathbf{BCDA}^{-1} - \mathbf{B}(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1} \\ &\quad - \mathbf{BCDA}^{-1}\mathbf{B}(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1} \\ &= \mathbf{I} + \mathbf{BCDA}^{-1} - [\mathbf{B} + \mathbf{BCDA}^{-1}\mathbf{B}][\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1}. \end{aligned}$$

If we subtract $\mathbf{I}$ from both sides and factor $\mathbf{BC}$ from the left on the third term, we find

$$\mathbf{0} = \mathbf{BCDA}^{-1} - \mathbf{BC}[\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B}][\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1}.$$

1 If $\mathbf{A}^{ij}$ is the $n-1 \times n-1$ matrix (minor) found by deleting row i and column j from $\mathbf{A}$, then the entry in row i and column j of the $\text{adj } \mathbf{A}$ is $(-1)^{i+j} \det \mathbf{A}^{ji}$.

which is

$$\mathbf{0} = \mathbf{0}, \quad \text{which was to be demonstrated.}$$

C.2 Eigenvalues and Eigenvectors

We consider the discrete dynamic system

$$\mathbf{x}_{k+1} = \Phi \mathbf{x}_k, \quad (\text{C.7})$$

where, for purposes of illustration, we will let

$$\Phi = \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ 1 & 0 \end{bmatrix}. \quad (\text{C.8})$$

If we assume that it is possible for this system to have a motion given by a geometric series such as z^k , we can assume that there is a vector $\mathbf{v}$ so that $\mathbf{x}_k$ can be written

$$\mathbf{x}_k = \mathbf{v} z^k. \quad (\text{C.9})$$

Substituting Eq. (C.9) into Eq. (C.7), we must find the vector $\mathbf{v}$ and the number z such that

$$\mathbf{v} z^{k+1} = \Phi \mathbf{v} z^k,$$

or, multiplying by z^{-k} yields

$$\mathbf{v} z = \Phi \mathbf{v}. \quad (\text{C.10})$$

If we collect both the terms of Eq. (C.10) on the left, we find

$$(z\mathbf{I} - \Phi)\mathbf{v} = \mathbf{0}. \quad (\text{C.11})$$

These linear equations have a solution for a nontrivial $\mathbf{v}$ if and only if the determinant of the coefficient matrix is zero. This determinant is a polynomial of degree n in z (Φ is an $n \times n$ matrix) called the **characteristic polynomial** of Φ , and values of z for which the characteristic polynomial is zero are roots of the characteristic equation and are called **eigenvalues** of Φ . For example, for the matrix given in Eq. (C.8) the characteristic polynomial is

$$\det \left\{ \begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} - \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ 1 & 0 \end{bmatrix} \right\}.$$

Adding the two matrices, we find

$$\det \begin{bmatrix} z - \frac{5}{6} & +\frac{1}{6} \\ -1 & z \end{bmatrix},$$

which can be evaluated to give

$$z(z - \frac{5}{6}) + \frac{1}{6} = (z - \frac{1}{2})(z - \frac{1}{3}). \quad (\text{C.12})$$

Thus the characteristic roots of this Φ are $\frac{1}{2}$ and $\frac{1}{3}$. Associated with these characteristic roots are solutions to Eq. (C.11) for vectors $\mathbf{v}$, called the **characteristic** or **eigenvectors**. If we let $z = \frac{1}{2}$, then Eq. (C.11) requires

$$\left\{ \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} - \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ 1 & 0 \end{bmatrix} \right\} \begin{bmatrix} v_{11} \\ v_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (\text{C.13})$$

Adding the matrices, we find that these equations become

$$\begin{bmatrix} -\frac{1}{3} & \frac{1}{6} \\ -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (\text{C.14})$$

Equations (C.14) are satisfied by any v_{11} and v_{21} such that

$$v_{21} = 2v_{11},$$

from which we conclude that the eigenvector corresponding to $z_1 = 1/2$ is given by

$$\mathbf{v}_1 = \begin{bmatrix} a \\ 2a \end{bmatrix}. \quad (\text{C.15})$$

We can arbitrarily select the scale factor a in Eq. (C.15). Some prefer to make the length² of eigenvectors equal to one. Here we make the largest component of $\mathbf{v}$ have unit magnitude. Thus the scaled $\mathbf{v}_1$ is

$$\mathbf{v}_1 = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}. \quad (\text{C.16})$$

In similar fashion, the eigenvector $\mathbf{v}_2$ associated with $z_2 = 1/3$ can be computed to be

$$\mathbf{v}_2 = \begin{bmatrix} \frac{1}{3} \\ 1 \end{bmatrix}.$$

Note that even if all elements of Φ are real, it is possible for characteristic values and characteristic vectors to be complex.

² Usually we define the length of a vector as the square root of the sum of squares of its components or, if $||\mathbf{v}||$ is the symbol for length, then $||\mathbf{v}||^2 = \mathbf{v}^T \mathbf{v}$. If $\mathbf{v}$ is complex, as will happen if z_i is complex, then we must take a conjugate, and we define $||\mathbf{v}||^2 = (\mathbf{v}^*)^T \mathbf{v}$, where $\mathbf{v}^*$ is the complex conjugate of $\mathbf{v}$.

C.3 Similarity Transformations

If we make a change of variables in Eq. (C.7) according to $\mathbf{x} = \mathbf{T}\mathbf{x}$, where $\mathbf{T}$ is an $n \times n$ matrix, then we start with the equations

$$\mathbf{x}_{k+1} = \Phi \mathbf{x}_k$$

and, substituting for $\mathbf{x}$, we have

$$\mathbf{T}\mathbf{x}_{k+1} = \Phi \mathbf{T}\mathbf{x}_k.$$

Then, if we multiply on the left by $\mathbf{T}^{-1}$, we get the equation in $\mathbf{x}$,

$$\mathbf{x}_{k+1} = \mathbf{T}^{-1}\Phi \mathbf{T}\mathbf{x}_k. \quad (\text{C.17})$$

If we define the new system matrix as Ψ , then the new states satisfy the equations

$$\mathbf{x}_{k+1} = \Psi \mathbf{x}_k,$$

where

$$\Psi = \mathbf{T}^{-1}\Phi \mathbf{T}. \quad (\text{C.18})$$

If we now seek the characteristic polynomial of Ψ , we find

$$\det[z\mathbf{I} - \Psi] = \det[z\mathbf{I} - \mathbf{T}^{-1}\Phi \mathbf{T}].$$

Because $\mathbf{T}^{-1}\mathbf{T} = \mathbf{I}$, we can write this polynomial as

$$\det[z\mathbf{T}^{-1}\mathbf{T} - \mathbf{T}^{-1}\Phi \mathbf{T}],$$

and the $\mathbf{T}^{-1}$ and $\mathbf{T}$ can be factored out on the left and right to give

$$\det[\mathbf{T}^{-1}[z\mathbf{I} - \Phi]\mathbf{T}].$$

Now, using property Eq. (C.1) for the determinant, we compute

$$\det \mathbf{T}^{-1} \cdot \det [z\mathbf{I} - \Phi] \cdot \det \mathbf{T},$$

which, by the equation following Eq. (C.4), gives us the final result

$$\det [z\mathbf{I} - \Psi] = \det [z\mathbf{I} - \Phi]. \quad (\text{C.19})$$

From Eq. (C.19) we see that Ψ and Φ have the same characteristic polynomials. The matrices are said to be “similar,” and the transformation Eq. (C.18) is a similarity transformation.

A case of a similarity transformation of particular interest is one for which the resulting matrix Ψ is diagonal. As an attempt to find such a matrix, suppose we *assume* that Ψ is diagonal and write the transformation $\mathbf{T}$ in terms of its columns, $\mathbf{t}_i$. Then Eq. (C.18) can be expressed as

$$\begin{aligned} \mathbf{T}\Psi &= \Phi \mathbf{T}, \\ [\mathbf{t}_1 \ \mathbf{t}_2 \ \dots \ \mathbf{t}_n] \Psi &= \Phi [\mathbf{t}_1 \ \mathbf{t}_2 \ \dots \ \mathbf{t}_n] \\ &= [\Phi \mathbf{t}_1 \ \Phi \mathbf{t}_2 \ \dots \ \Phi \mathbf{t}_n]. \end{aligned} \quad (\text{C.20})$$

If we *assume* that Ψ is diagonal with elements $\lambda_1, \lambda_2, \dots, \lambda_n$, then Eq. (C.20) can be written as

$$[\mathbf{t}_1 \ \mathbf{t}_2 \ \dots \ \mathbf{t}_n] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ 0 & & \ddots & \\ \vdots & & & \lambda_n \end{bmatrix} = [\Phi \mathbf{t}_1 \ \Phi \mathbf{t}_2 \ \dots \ \Phi \mathbf{t}_n].$$

Multiplying the matrices on the left, we find

$$[\lambda_1 \mathbf{t}_1 \ \lambda_2 \mathbf{t}_2 \ \dots \ \lambda_n \mathbf{t}_n] = [\Phi \mathbf{t}_1 \ \dots \ \Phi \mathbf{t}_n]. \quad (\text{C.21})$$

Because the two sides of Eq. (C.21) are equal, they must match up column by column, and we can write the equation for column j as

$$\lambda_j \mathbf{t}_j = \Phi \mathbf{t}_j. \quad (\text{C.22})$$

Comparing Eq. (C.22) with Eq. (C.10), we see that $\mathbf{t}_j$ is an eigenvector of Φ and λ_j is an eigenvalue. We conclude that if the transformation $\mathbf{T}$ converts Φ into a diagonal matrix Ψ , then the columns of $\mathbf{T}$ must be eigenvectors of Φ and the diagonal elements of Ψ are the eigenvalues of Φ [which are also the eigenvalues of Ψ , by Eq. (C.19)]. It turns out that if the eigenvalues of Φ are distinct, then there are exactly n eigenvectors and they are independent; that is, we can construct a nonsingular transformation $\mathbf{T}$ from the n eigenvectors.

In the example given above, we would have

$$\mathbf{T} = \begin{bmatrix} \frac{1}{2} & \frac{1}{3} \\ 1 & 1 \end{bmatrix},$$

for which

$$\mathbf{T}^{-1} = \begin{bmatrix} 6 & -2 \\ -6 & 3 \end{bmatrix},$$

and the new diagonal system matrix is

$$\begin{aligned} \mathbf{T}^{-1} \Phi \mathbf{T} &= \begin{bmatrix} 6 & -2 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{3} \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{3} \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \end{aligned}$$

as advertised! If the elements of Φ are real and an eigenvalue is complex, say $\lambda_1 = \alpha + j\beta$, then the conjugate, $\lambda_1^* = \alpha - j\beta$, is also an eigenvalue because the characteristic polynomial has real coefficients. In such a case, the respective eigenvectors will be conjugate. If $\mathbf{v}_1 = \mathbf{r} + j\mathbf{i}$, then $\mathbf{v}_2 = \mathbf{v}_1^* = \mathbf{r} - j\mathbf{i}$, where $\mathbf{r}$

and $\mathbf{i}$ are matrices of real elements representing the real and imaginary parts of the eigenvectors. In such cases, it is common practice to use the real matrices $\mathbf{r}$ and $-\mathbf{i}$ as columns of the transformation matrix $\mathbf{T}$ rather than go through the complex arithmetic required to deal directly with $\mathbf{v}_1$ and $\mathbf{v}_1^*$. The resulting transformed equations are not diagonal, but rather the corresponding variables appear in the coupled equations

$$\dot{\eta} = \alpha\eta - \beta v, \quad \dot{v} = \beta\eta + \alpha v. \quad (\text{C.23})$$

C.4 The Cayley-Hamilton Theorem

A very useful property of a matrix Φ follows from consideration of the inverse of $z\mathbf{I} - \Phi$. As we saw in Eq. (C.5), we can write

$$(z\mathbf{I} - \Phi) \text{adj}(z\mathbf{I} - \Phi) = \mathbf{I} \det(z\mathbf{I} - \Phi). \quad (\text{C.24})$$

The coefficient of $\mathbf{I}$ on the right-hand side of Eq. (C.24) is the characteristic polynomial of Φ , which we can write as

$$a(z) = z^n + a_1 z^{n-1} + a_2 z^{n-2} + \cdots + a_n.$$

The adjugate of $z\mathbf{I} - \Phi$, on the other hand, is a *matrix* of polynomials in z , found from the determinants of the minors of $z\mathbf{I} - \Phi$. If we collect the constant matrix coefficients of the powers of z , it is clear that we can write

$$\text{adj}(z\mathbf{I} - \Phi) = \mathbf{B}_1 z^{n-1} + \mathbf{B}_2 z^{n-2} + \cdots + \mathbf{B}_n,$$

and Eq. (C.24) becomes a polynomial equation with matrix coefficients. Written out, it is

$$[z\mathbf{I} - \Phi][\mathbf{B}_1 z^{n-1} + \mathbf{B}_2 z^{n-2} + \cdots + \mathbf{B}_n] = z^n \mathbf{I} + a_1 \mathbf{I} z^{n-1} + \cdots + a_n \mathbf{I}. \quad (\text{C.25})$$

If we now multiply the two matrices on the left and equate coefficients of equal powers of z , we find

$$\begin{aligned} \mathbf{B}_1 &= \mathbf{I}, \\ \mathbf{B}_2 &= \Phi \mathbf{B}_1 + a_1 \mathbf{I} = \Phi + a_1 \mathbf{I}, \\ \mathbf{B}_3 &= \Phi \mathbf{B}_2 + a_2 \mathbf{I} = \Phi^2 + a_1 \Phi + a_2 \mathbf{I}, \\ &\vdots \\ \mathbf{B}_n &= \Phi \mathbf{B}_{n-1} + a_{n-1} \mathbf{I} = \Phi^{n-1} + a_1 \Phi^{n-2} + \cdots + a_{n-1} \mathbf{I}, \\ 0 &= \Phi \mathbf{B}_n + a_n \mathbf{I} = \Phi^n + a_1 \Phi^{n-1} + a_2 \Phi^{n-2} + \cdots + a_n \mathbf{I}. \end{aligned} \quad (\text{C.26})$$

Equation (C.26) is a statement that the matrix obtained when matrix Φ is substituted for z in the characteristic polynomial, $a(z)$, is exactly zero! In other words, we have the Cayley-Hamilton theorem according to which

$$a(\Phi) = 0. \quad (\text{C.27})$$

• D •

Summary of Facts from the Theory of Probability and Stochastic Processes

D.1 Random Variables

We begin with a space of experiments, Ω , whose outcomes are called ω and depend on chance. Over the space Ω and its subsets Ω_i we define a probability function P ,¹ which assigns a positive number between 0 and 1 to each countable combination of subsets in Ω to which an outcome (or “event”) can belong. P has the properties that the probability of *some* outcome is certain, in which case it is assigned the value 1

$$P(\Omega) = 1,$$

and the probability of an outcome that may result from events Ω_i, Ω_j , which have no common points (the intersection of Ω_i and Ω_j is empty) is the sum of the probabilities of Ω_i and Ω_j

$$P\{\Omega_i \cup \Omega_j\} = P(\Omega_i) + P(\Omega_j).$$

In addition to the function P we define a **random variable** $x(\omega)$, which maps Ω into the real line such that to each outcome ω in Ω we associate a value x , and the probability that a chance experiment maps into a value x that is less than or equal to the constant a is²

$$\Pr(x \leq a) = F_x(a). \quad (\text{D.1})$$

¹ See Parzen (1962).

² $\Pr \{ \cdot \}$ is meant to be read “the probability that $\{ \cdot \}$.”

The function $F_x(\xi)$ is called the **distribution function** of the random variable. If F_x is a smooth function,³ we define its derivative $f_x(\xi)$ as the **density function**, which has the property

$$\Pr\{a \leq x \leq b\} = \int_a^b f_x(\xi) d\xi. \quad (\text{D.2})$$

Because the whole space Ω maps into the line somewhere, we have

$$\int_{-\infty}^{\infty} f_x(\xi) d\xi = 1. \quad (\text{D.3})$$

Two common density functions that we shall have reason to use are the **uniform density** and the **normal** or **Gaussian density**. A random variable having a uniform density has zero probability of having any value outside a finite range between lower limit ℓ and upper limit u , $\ell \leq x \leq u$, and f_x is constant inside this range. Because of Eq. (D.3), the constant is $1/(u - \ell)$. A sketch of the uniform density is given in Fig. D.1.

Figure D.1
Sketch of the uniform
density function

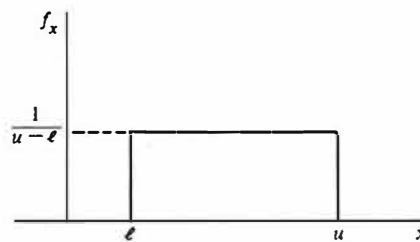
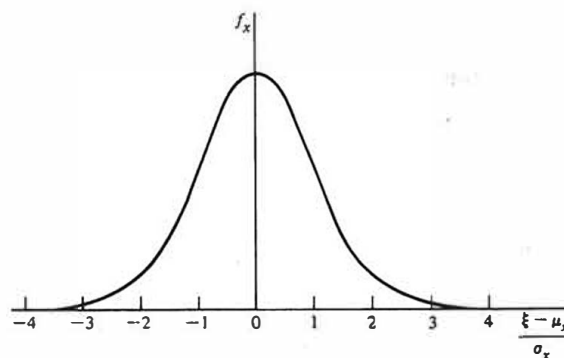


Figure D.2
Sketch of the normal
density function



³ Using the impulse, we can include simple discontinuities, as must be done if a specific value of x has nonzero probability.

The normal density function is given by the equation

$$f_x(\xi) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{1}{2} \frac{(\xi - \mu_x)^2}{\sigma_x^2}\right) \quad (\text{D.4})$$

and shown in the sketch in Fig. D.2.

The importance of the normal density derives mainly from the following facts:

1. The distribution of a random variable based on events which themselves consist of a sum of a large number of independent⁴ random events is accurately approximated by the normal law. Such a distribution describes electrical noise caused by thermal motions of a large number of particles as in a resistor, for example.
2. If two random variables have (jointly) normal distributions, then their sum also has a normal distribution. (As an extension of this second point, if the input to a linear system is normal, then the distribution of its output is also normal.)

D.2 Expectation

By the very nature of variables whose values are dependent on chance, we cannot discuss a formula for the calculation of values of x . To describe a random variable, we instead discuss average values such as the arithmetic mean or the average power. Such concepts are contained in the idea of **expectation**. The expected value of a function g of a random variable whose density is f_x is defined as

$$\begin{aligned} \mathcal{E}\{g(x)\} &= \int_{-\infty}^{\infty} g(\xi) f_x(\xi) d\xi \\ &= \overline{g(x)}. \end{aligned} \quad (\text{D.5})$$

Important special cases are the mean, variance, and mean square. If $g(x) = x$, then we have the mean, namely

$$\mathcal{E}(x) = \int_{-\infty}^{\infty} \xi f_x(\xi) d\xi = \bar{x} = \mu_x. \quad (\text{D.6})$$

For the uniform distribution, the mean is given by

$$\begin{aligned} \bar{x} &= \int_{-\infty}^{\infty} \xi f_x(\xi) d\xi \\ &= \int_{\ell}^u \left[\frac{1}{u - \ell} \right] d\xi = \frac{1}{u - \ell} \left[\frac{\xi^2}{2} \right]_{\ell}^u \end{aligned}$$

⁴ The technical definition of "independent" will be given shortly.

$$= \frac{1}{u - \ell} \left[\frac{u^2}{2} - \frac{\ell^2}{2} \right] = \frac{u + \ell}{2}. \quad (\text{D.7})$$

Because the probability density function has the intuitive properties of a histogram of relative frequency of occurrence of a particular x , we see that the mean, like an arithmetic mean, is a weighted average of the random variable values.

If $g(x)$ in Eq. (D.5) is $g = (x - \bar{x})^2$, then the expected value is the average of the square of the variation of the variable from the mean; this number is called the **variance** of x , written $\text{var } x$. For the uniform density we compute

$$\begin{aligned} \mathcal{E}(x - \bar{x})^2 &= \text{var } x \\ &= \int_{-\infty}^{\infty} (\xi - \bar{x})^2 f_x(\xi) d\xi \\ &= \int_{\ell}^u (\xi - \bar{x})^2 \frac{1}{u - \ell} d\xi \\ &= \frac{1}{u - \ell} \left[\frac{(\xi - \bar{x})^3}{3} \right]_{\ell}^u \\ &= \frac{1}{u - \ell} \left\{ \frac{(u - \bar{x})^3}{3} - \frac{(\ell - \bar{x})^3}{3} \right\}. \end{aligned}$$

Substituting Eq. (D.7) for $\bar{x}$, we find

$$\mathcal{E}(x - \bar{x})^2 = \frac{1}{3(u - \ell)} \left\{ \left(\frac{u - \ell}{2} \right)^3 - \left(\frac{\ell - u}{2} \right)^3 \right\},$$

and, simplifying, we get

$$\text{var } x = \frac{(u - \ell)^2}{12}. \quad (\text{D.8})$$

The square root of the variance is called the **standard deviation** and given the symbol σ . Thus, for the uniform density

$$\sigma_x^2 = \frac{(u - \ell)^2}{12}. \quad (\text{D.9})$$

If we let $g(x)$ be simply x^2 , then we compute by Eq. (D.5) the mean-square value $\overline{x^2}$. However, consider an expanded expression for the variance, and let the mean value of x be μ

$$\begin{aligned} \text{var } x &= \int_{-\infty}^{\infty} (\xi - \mu)^2 f_x(\xi) d\xi \\ &= \int_{-\infty}^{\infty} (\xi^2 - 2\xi\mu + \mu^2) f_x(\xi) d\xi \\ &= \overline{x^2} - 2\mu^2 + \mu^2 \\ &= \overline{x^2} - \mu^2. \end{aligned}$$

Thus we find

$$\overline{x^2} = \text{var } x + \mu^2.$$

For the uniform density, the mean-square value is, therefore

$$\overline{x^2} = \frac{(u - \ell)^2}{12} + \left(\frac{u + \ell}{2} \right)^2.$$

The integration is more complicated in the case of the normal density, but the definitions are the same; namely, if x has the normal density given by Eq. (D.4), then

$$\bar{x} = \int_{-\infty}^{\infty} \xi f_x(\xi) d\xi = \mu_x \quad (\text{D.10a})$$

and

$$\begin{aligned} \mathcal{E}(x - \mu_x)^2 &= \int_{-\infty}^{\infty} (\xi - \mu_x)^2 f_x(\xi) d\xi \\ \text{var } x &= \sigma_x^2. \end{aligned} \quad (\text{D.10b})$$

Note that the mean and standard deviation already appear as parameters of the density given by Eq. (D.4). Because these two parameters completely describe the normal density, it is standard to say that a random variable with normal density having mean μ and standard deviation σ is distributed according to the $N(\mu, \sigma)$ law.

D.3 More Than One Random Variable

Frequently the random experiment has an outcome that is mapped into several random variables, say $x_1, x_2, \dots, x_n$ or, organized as a column matrix

$$\mathbf{x} = (x_1 x_2 \dots x_n)^T.$$

For this collection we define the probability that

$$x_1 \leq a_1, x_2 \leq a_2, \dots, x_n \leq a_n$$

as

$$\Pr\{\mathbf{x} \leq \mathbf{a}\}$$

and let this be the distribution $F_x(\xi)$ with vector argument. The corresponding density $f_x(\xi)$ is a function such that the probability that $\mathbf{x}$ is in a box with sides a_i and b_i is given by

$$\Pr\{\mathbf{a} < \mathbf{x} \leq \mathbf{b}\} = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} f_x(\xi_1, \dots, \xi_n) d\xi_1 \dots d\xi_n. \quad (\text{D.11})$$

The mean of the vector is a vector of the means

$$\mathcal{E}\{\mathbf{x}\} = \boldsymbol{\mu}_x = (\mu_{x_1} \mu_{x_2} \dots \mu_{x_n})^T, \quad (\text{D.12})$$

and likewise the deviation of the random variables from the mean-value vector is measured by a matrix of terms called the **covariance matrix**, $\text{cov } \mathbf{x}$, defined as

$$\text{cov } \mathbf{x} = \mathcal{E}\{(\mathbf{x} - \boldsymbol{\mu}_x)(\mathbf{x} - \boldsymbol{\mu}_x)^T\} = \mathbf{R}_{xx}. \quad (\text{D.13})$$

Thus $\text{cov } \mathbf{x}$ is a matrix with the element in row i and column j given by

$$\mathcal{E}\{(x_i - \mu_{x_i})(x_j - \mu_{x_j})\}.$$

Often the symbol $\mathbf{R}_x$ will be used for the covariance matrix of a random vector $\mathbf{x}$. In an obvious way we can extend Eq. (D.13) to include the case of different random vectors $\mathbf{x}$ and $\mathbf{y}$ with mean vectors $\boldsymbol{\mu}_x$ and $\boldsymbol{\mu}_y$. We define the covariance of $\mathbf{x}$ and $\mathbf{y}$ as

$$\text{cov}(\mathbf{x}, \mathbf{y}) = \mathcal{E}\{(\mathbf{x} - \boldsymbol{\mu}_x)(\mathbf{y} - \boldsymbol{\mu}_y)^T\} = \mathbf{R}_{xy}. \quad (\text{D.14})$$

Consider now the special case of two random variables x and y having distribution $F_{x,y}(\xi, \eta)$ and (joint) density $f_{x,y}(\xi, \eta)$. Because the probability that y has any value is the integral from $-\infty$ to $+\infty$ over η , the density of x is given by

$$f_x(\xi) = \int_{-\infty}^{\infty} f_{x,y}(\xi, \eta) d\eta. \quad (\text{D.15})$$

If $f_{x,y}$ is such that

$$f_{x,y}(\xi, \eta) = f_x(\xi) f_y(\eta), \quad (\text{D.16})$$

we say that x and y are **independent**.

It often happens that we know a particular value of one random variable, such as the output y of a dynamic system, and we wish to estimate the value of a related variable, such as the state x . A useful function for this situation is the **conditional density**, defined as the density of x given that the value of y is η and is defined as

$$f_{x|y}(\xi | \eta) = \frac{f_{xy}(\xi, \eta)}{f_y(\eta)}. \quad (\text{D.17})$$

Note that, from Eqs. (D.15) and (D.16), if x and y are independent, then

$$f_{x|y}(\xi | \eta) = f_x(\xi). \quad (\text{D.18})$$

The most important multivariable probability density is the normal or Gaussian law given by

$$f_x(\xi) = [(2\pi)^n |\mathbf{R}_x|]^{-1/2} \exp \left[-\frac{1}{2} (\xi - \boldsymbol{\mu}_x)^T \mathbf{R}_x^{-1} (\xi - \boldsymbol{\mu}_x) \right], \quad (\text{D.19})$$

where $|\mathbf{R}_x|$ is the determinant of the matrix $\mathbf{R}_x$.